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Freezing Shrinkage Dynamics and Surface Dendritic Growth of Floating Refractory Alloy Droplets in Outer Space

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The freezing shrinkage and dendritic growth are of great importance for various alloys solidified from high-temperature liquids to solids since they dominate microstructure patterns and follow-up processing. However, the microgravity freezing shrinkage dynamics is scarcely explored on the ground as it is hard to suppress the strong natural convection inside liquid alloys. Here, a series of in-orbit solidification experiments is conducted aboard the China Space Station with a long-term stable 10^{-5} g_0 microgravity condition. The highest temperature up to 2265 K together with substantial liquid undercoolings far from a thermodynamically stable state are attained for both $Nb_{82.7}Si_{17.3}$ and $Zr_{64}V_{36}$ refractory alloys. Furthermore, the solidification under microgravity of a droplet is simulated to reveal the liquid–solid interface migration, temperature gradient, and flow field. The microgravity solidification process leads to freezing shrinkage cavities and distinctive surface dendritic microstructure patterns. The combined effects of shrinkage dynamics and liquid surface flow in outer space result in the dendrites growing not only along the tangential direction but also along the normal direction to the droplet surface. These space experimental results contribute to a further understanding of the solidification behavior of liquid alloys under a weaker convection condition, which is often masked by gravity on the ground.

1. Introduction

The liquid–solid phase transition has a strong relationship between the microstructure/performance and the physical fields of solidification, then resulting in the deep undercooling, rapid solidification dynamics, and multiple microstructure patterns.[1–8] In outer space where a stable microgravity condition is provided, the solidification of liquid alloy would not only produce new microstructures but also reveal various interesting scientific puzzles concealed by the earth's gravity. However, there are still many challenges to perform this kind of experiment due to the scarce chances and high temperatures, especially for refractory alloys with liquidus temperature beyond 2000 K. Therefore, various methods were used to achieve the microgravity environment, measure thermophysical properties, obtain novel microstructures, and promote the development of solidification theories.[5,9–11]

China Space Station (CSS) was completely set up in November 2022, with a microgravity level of 10^{-5} g_0 (g_0 is the acceleration of gravity on the ground). The buoyancy convection caused by gravity in the solidification process of the high-temperature droplets can be ignored at such a microgravity level. Both the International Space Station (ISS) and CSS are supplied with experimental equipment to study the alloy solidification under microgravity and containerless conditions.[12,13] Electrostatic levitation (ESL) has been widely used in solidification experiments, most of which are carried out by ground-based ESL equipment,[14–17] while a small number of which are carried out on ISS.[18] Compared to electromagnetic levitation, ESL lacks electromagnetic stirring within the molten materials, which is advantageous for studying deep undercooling and rapid solidification of liquid alloys. ESL facilitates more precise measurements of thermophysical properties since it is more stable compared to acoustic levitation. Ground-based ESL needs a strong electrostatic field to counteract gravity, while space-based ESL only provides a weaker electrostatic field to control sample position. Recently, the experimental results of immiscible alloy conducted aboard CSS have also been reported to reveal the phase separation in outer space.[19] But for refractory alloys, the dendritic and eutectic growth on the surface or inside the levitated droplet is scarcely investigated. We care about the questions as follows: a) Could an undercooled state deviating from thermodynamical stability be achieved for the refractory alloys in outer space? b) Could the dendrites/eutectics grow not only along the tangential direction but also along the normal direction to the droplet surface? c) Could the microgravity experiments produce novel microstructure patterns?

The Nb—Si and Zr—V alloys are typical refractory alloys, with Nb—Si alloy holding great potential to be applied to the ultrahigh-temperature aviation engine and Zr—V alloy showing strong prospects in nuclear materials applications. These two alloys are the typical refractory materials with dendritic and eutectic growth

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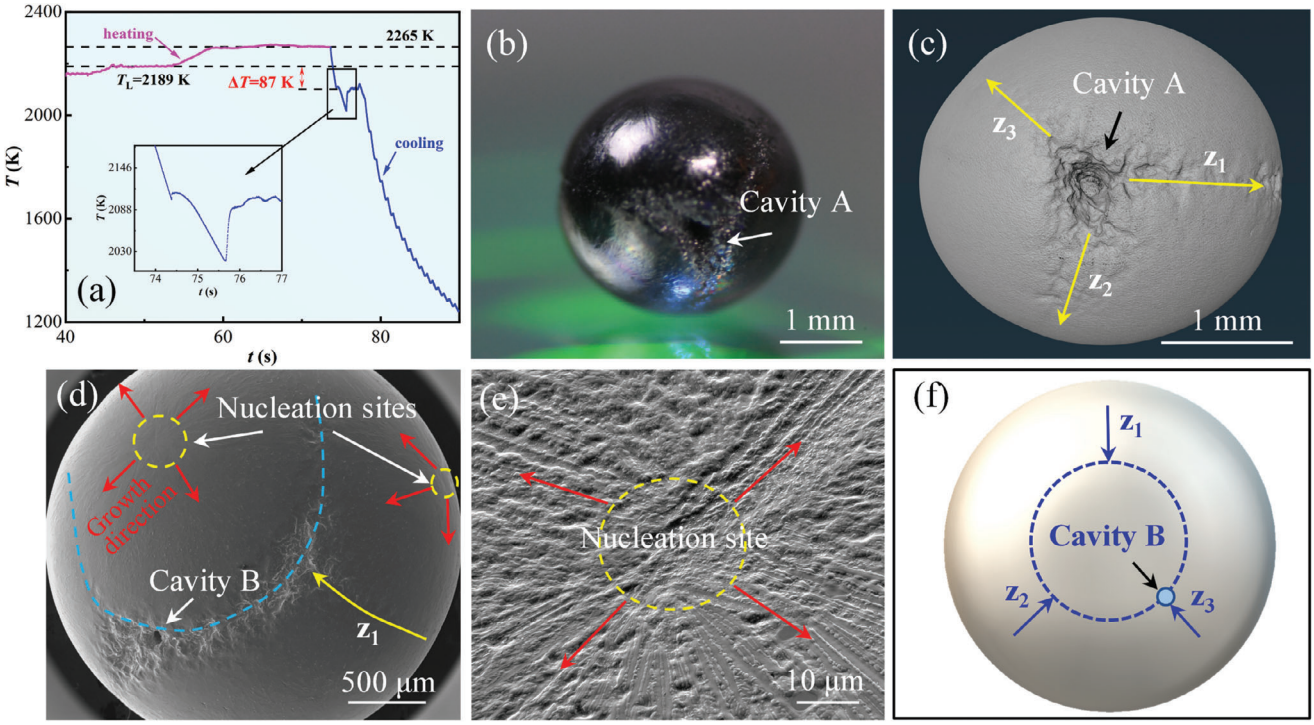

Figure 1. Surface morphology of microgravity freezing sphere. a) Temperature curve of $Nb_{82.7}Si_{17.3}$ alloy droplet with 87 K undercooling. b) Solidified droplet morphology around shrinkage cavity A. c) Morphology around shrinkage cavity A. d) Morphology around shrinkage cavity B. e) Microstructure around the nucleation site. f) Sketch of surface morphology.

and display a competence growth between the primary solution solid phase and the eutectic phase.[20,21] Especially, the eutectic growth of these two alloys involves a solid solution ((Nb) or (Zr)) and an intermetallic compound (Nb_3Si or V_2Zr). For this kind of complicated phase transition, it is hard to clearly clarify the experimented growth manner on the ground. It is necessary to perform such a fundamental study on microgravity solidification dynamics and microstructure patterns. The aim of this article is to elucidate the freezing shrinkage dynamics and surface dendritic growth of floating eutectic $Nb_{82.7}Si_{17.3}$ and hypoeutectic $Zr_{64}V_{36}$ refractory alloy droplets in outer space.

2. Results and Discussion

2.1. Freezing Shrinkage Dynamics Coupled with Microgravity Flow Mechanism

The surface of the eutectic $Nb_{82.7}Si_{17.3}$ alloy spheres have groove-like and hole-like shrinkage cavities. Meanwhile, the shrinkage cavities are also observed on the surface of the hypoeutectic $Zr_{64}V_{36}$ alloy spheres. However, the shrinkage cavities on the surface of hypoeutectic $Zr_{64}V_{36}$ alloy spheres are only holes but not grooves. Similar surface morphology was observed on those samples' surfaces if the undercooling is within a certain range.

The morphologies of eutectic $Nb_{82.7}Si_{17.3}$ alloy spheres were displayed in **Figure 1**, which was solidified at an undercooling of 87 K, as shown in Figure 1a. Figure 1b shows the photo with a shrinkage cavity (it is numbered as shrinkage cavity A), and Figure 1c shows the CT scan photograph of shrinkage cavity A. The depth of shrinkage cavity A can be estimated to be ≈1.3 mm, almost half the diameter of the sphere. Another shrinkage cavity (labeled as B) is formed at the opposite side of shrinkage cavity A on the surface, and the positions circled by the yellow dash line in the figure are nucleation sites (Figure 1d). As indicated by the red arrows in Figure 1d, when two growing crystal grains originated from two independent nucleation sites meet together, the solidification of residual liquids at the junctions causes the formation of a groove-like cavity. In a similar way, a hole-like cavity is formed in the situation of three or more nucleation regions. At the location of nucleation sites marked in Figure 1d, the microstructure grows radially outward (Figure 1e), which is consistent with the microstructure characteristics of nucleation sites. Accordingly, a sketch around shrinkage cavity B is shown in Figure 1f. Directions z_1, z_2, and z_3 have the same meaning in Figure 1c,d,f, which represent the extension directions of the groove-like shrinkage cavities on the sample's surface. The blue dash lines in Figure 1d,f represent the groove-like shrinkage cavities. We deduced that multiple nucleation sites occurred on the droplet surface, and the groove-like shrinkage cavities are formed at the junction of two nucleation regions while the hole-like shrinkage cavity is formed at the intersection of three (or more) nucleation regions, which results from the combination of microgravity field and solidification dynamics.

It is worth to explore the microgravity liquid behavior since many studies have shown that the nucleation is strongly related to the liquid structure and properties.[21–30] The temperature field, flow field, and phase transition within liquid eutectic $Nb_{82.7}Si_{17.3}$ alloy sphere under microgravity were simulated by finite element

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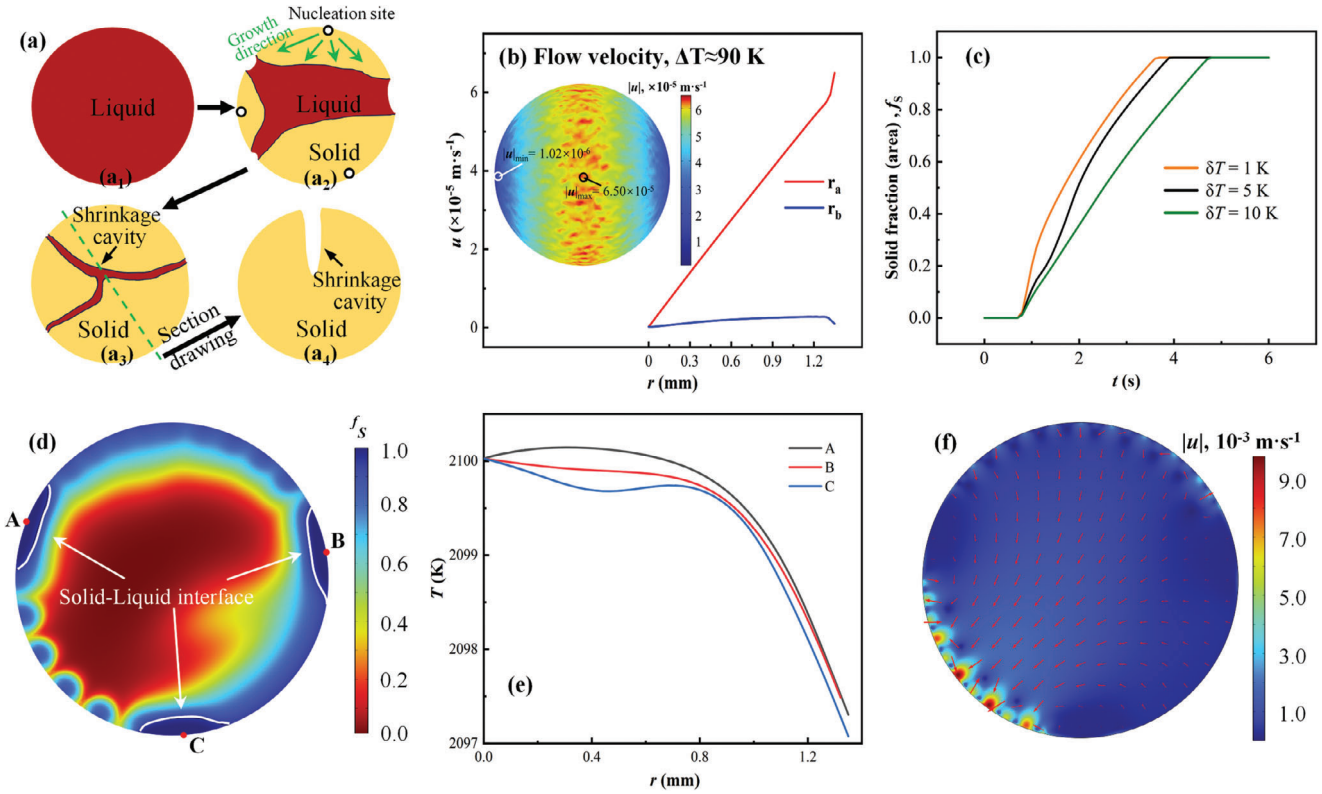

Figure 2. Shrinkage dynamics and microgravity flow of eutectic $Nb_{82.7}Si_{17.3}$ alloy. a) Sketch of shrinkage cavity formation. b) 3D flow velocity at 90 K undercooling. c) Solid fraction (area) versus time. d) 2D phase transition with multiple nucleation regions. e) Radial temperature distribution from droplet center to points A, B, and C designated in (d). f) Flow field corresponding to (d).

method (FEM) to reveal the shrinkage process (**Figure 2**). To establish an intuitive understanding, we first show the sketch of the shrinkage cavities formation mode with multiple nucleation sites of eutectic $Nb_{82.7}Si_{17.3}$ alloy (Figure 2a). When the distance between the solid regions is close, the liquid region between the solid will be concave inward (Figure $2a_2$), which is similar to the capillary effect. Once the solid regions contact each other and separate the residual liquid, it will lead to the appearance of some groove-like or hole-like shrinkage cavities with those separated liquid regions continue to solidify, for the solidification reduces the volume of the liquid region (Figure 2a,a_4). We established a 3D sphere model to reveal the flow velocity at an undercooling of 90 K (Figure 2b), which represents the real solidification temperature shown in Figure 1a. Figure 2b shows the flow velocity distribution, from which we can find that the maximum and the minimum flow velocity on the surface reach the order of 10^{-5} and 10^{-6} m s^{-1}, respectively. The simulated maximum flow velocity at the melting point in our model is consistent with Li's results.[31] The *r*-*u* curve describes the change of flow velocity along directions r_a and r_b (r_a represents the direction from the center of the droplet to the surface position where the flow velocity is the maximum, while r_b represents the direction which is the minimum). The solid fraction (area) changes with time at different temperature transition intervals δT are studied through a 2D model (Figure 2c). The solidification time increases with the increase of δT, and we choose the result of $\delta T = 3$ K to be used in the simulation because of its the closest solidification time to reality. Figure 2d–f show the typical phase transition, temperature, and flow field when there are multiple solid regions. The flow velocity is hundreds of times larger than that in Figure 2b during solidification, as shown in Figure 2f, reaching the order of 10^{-3} and 10^{-4} m s^{-1}. Meanwhile, we found that the flow restrains the solidification, which means that the solidification will preferentially move to areas with a lower flow velocity.

2.2. Surface Dendritic Growth under Microgravity State

In order to further reveal the dendritic microstructure, we take cavity A to clarify the crystal morphology on the surface (**Figure 3**). Figure 3a shows the overall morphology around shrinkage cavity A, and the interesting microstructures are marked by red boxes. Figure 3b,c show the primary dendrites growing parallel to the surface, where a single dendrite along the direction of the primary dendrite arm is shown (Figure 3b). Here, the Nb_3Si phase grows in a faceted pattern near the primary dendrites, which is highlighted in blue. This is because the composition of the residual liquid is far from the eutectic, which results in the growth of Nb_3Si phases. We find a special area around shrinkage cavity A, which infers an individual nucleation site here (Figure 3c). In Figure 3d, the extraordinary dendrites can be seen at the edge of shrinkage cavity A, which are in accordance with the preferred orientation relation of the crystallography: the preferred growth orientation of the body-centered cubic

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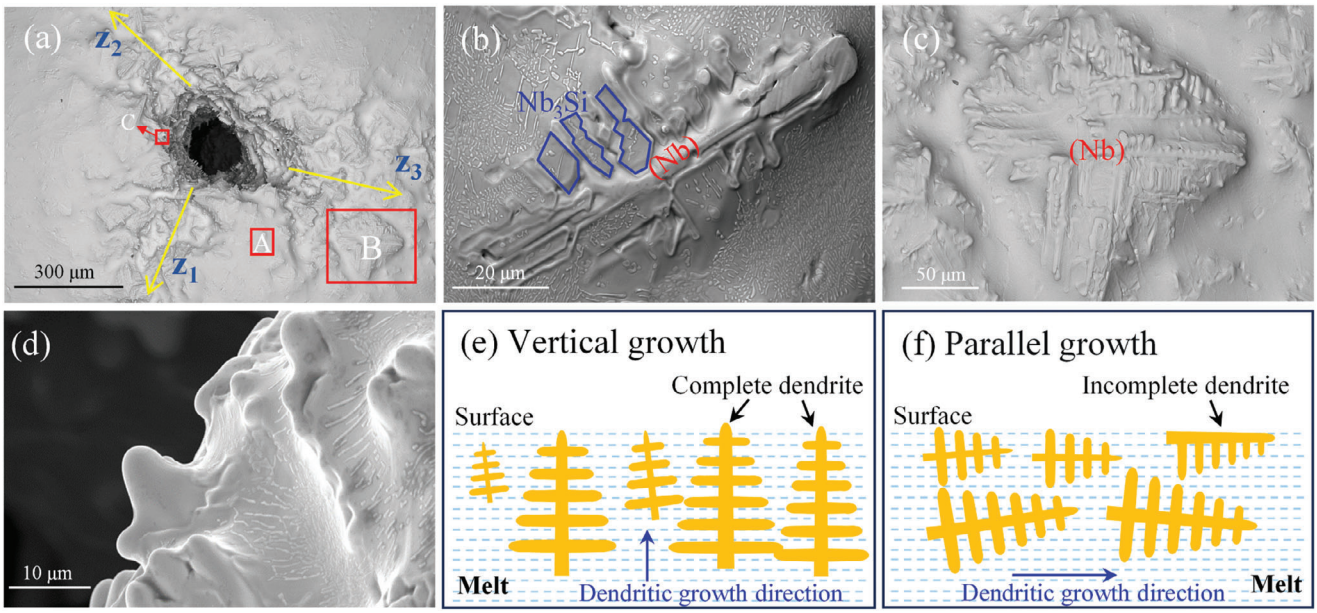

Figure 3. Surface dendritic microstructure of $Nb_{82.7}Si_{17.3}$ alloy. a) Overall morphology around shrinkage cavity A. b,c) Dendrites grow parallel to droplet surface at zones A and B of (a). d) Dendrites grow perpendicular to the droplet surface at zone C of (a). e,f) Sketches of vertical and parallel dendrite growth modes.

(BCC) lattice structure of primary (Nb) phases is <100>. This results in the status that the growth direction of the secondary dendrite arms is perpendicular to the cross shape of the primary arm, which is consistent with the result obtained by the phase-field crystal model.[32,33] As shown in the sketches of dendrite growth in the tangential and normal directions to the droplet surface in Figure 3e,f, the corresponding growth patterns can be found on the surface of $Nb_{82.7}Si_{17.3}$ alloy. These special dendrites result from just a surface effect under microgravity.

We found various interesting dendritic microstructures when we examined the characteristics along the inner wall of shrinkage cavity A. **Figure** 4a shows the overall morphology inside shrinkage cavity A and the typical microstructures are marked by the red box. Figure 4b shows some primary dendrites growing along different directions in the inner wall. The dendrites appearing in the hole-like cavity exhibit more developed secondary arms (Figure 4c,d) when compared to those appearing on the surface (Figure 3c,d). This is because the former possesses

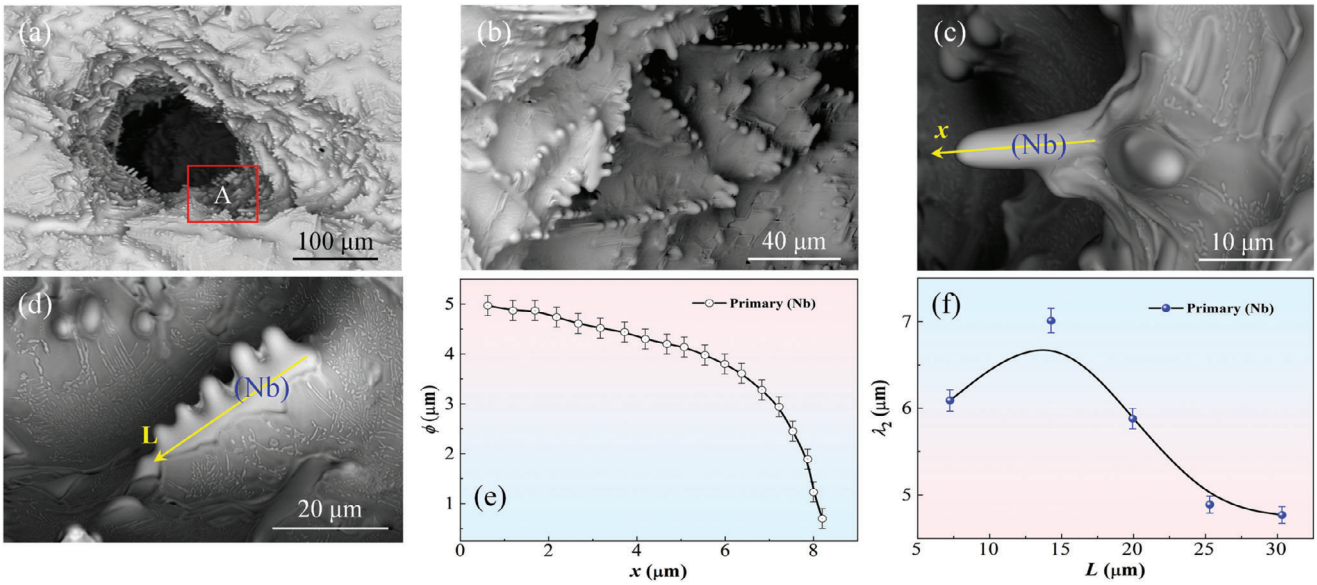

Figure 4. Dendritic morphology inside shrinkage cavity. a) Overall morphology inside shrinkage cavity A of eutectic $Nb_{82.7}Si_{17.3}$ alloy. b) Dendrites grow crisscross the inner wall at zone A of (a). c,d) Secondary dendrite arms grow perpendicular to the inner wall. e) Diameter of a (Nb) dendrite branch along the x direction in (c). f) Secondary dendrite arm spacing of the primary (Nb) phase along the L direction in (d).

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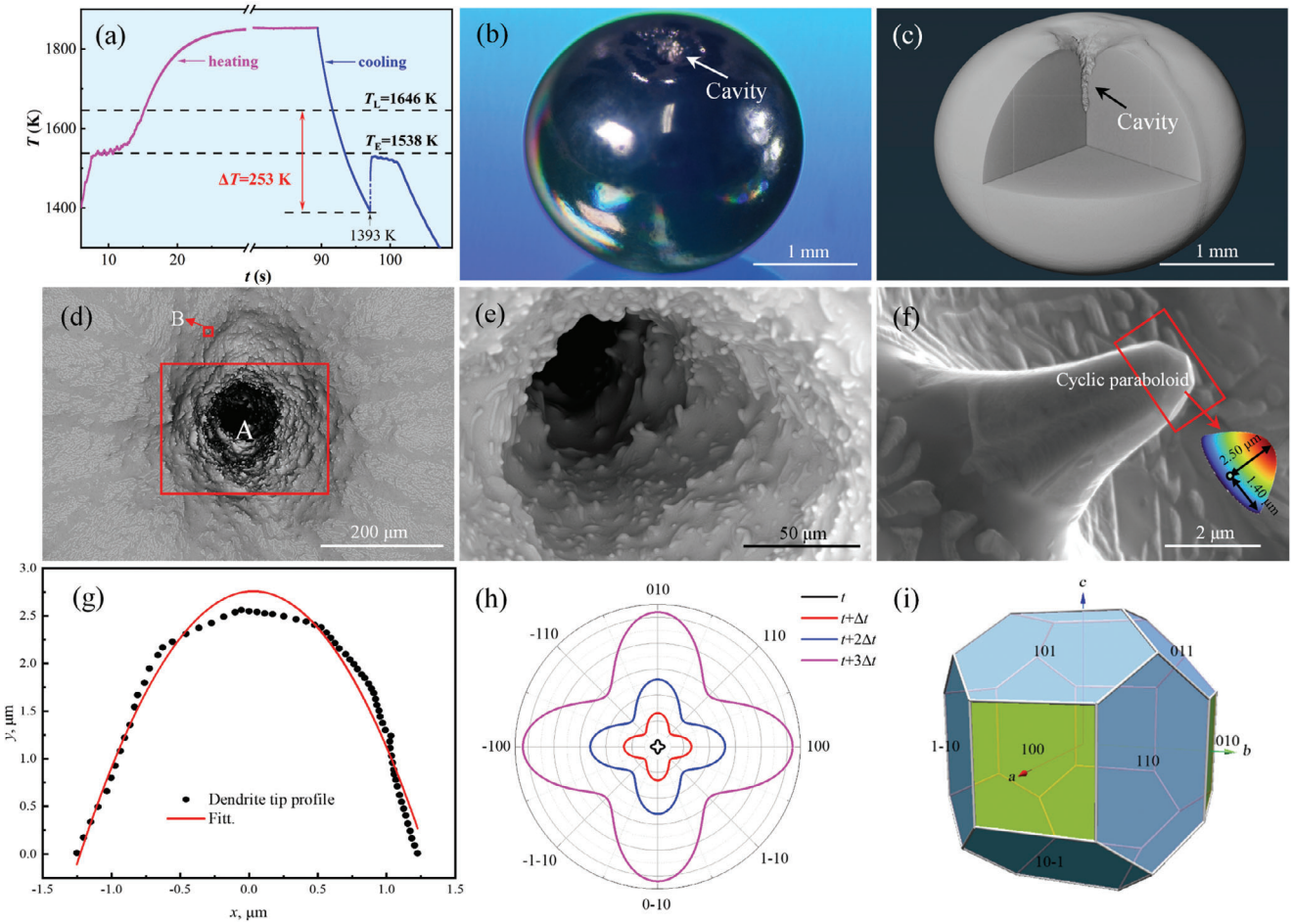

Figure 5. Shrinkage cavity morphology and surface dendritic microstructure of $Zr_{64}V_{36}$ alloy. a) Temperature curve with 253 K undercooling. b) Overall morphology of the solidified droplet. c) Internal and surface morphology of the alloy sphere. d) Overall morphology of the shrinkage cavity. e,f) Dendritic microstructures on the inner wall at zones A and B in (d). g) Dendrite tip profile of (Zr) phase in (f). h) Cross section profile of (Zr) dendrite during growth. i) Preferential growth direction of (Zr) phase.

a longer growth duration due to the decrease of undercooling within residual liquids, where the temperature on the droplet surface is lower than that in the hole (Figure 2e). The secondary arms of primary dendrites are longer compared to those in Figure 3d, ≈10 μm (Figure 4c,d). The dendrite branch diameter is measured to be 5 μm and exhibits a gradual decrease along the x direction (Figure 4e). Additionally, the secondary dendrite spacing λ_2 of the primary (Nb) phase along the L direction in Figure 4d measures ≈5–7 μm (Figure 4f). The decreasing λ_2 indicates that the dendrite grows along the L direction.

To recapitulate briefly, the primary (Nb) dendrites in the groove-like zone (Figure 3c) and hole-like zone (Figure 4b) both exhibit well-developed branches. Nevertheless, the difference lies in the growth direction, that is, dendrites near the groove-like zone grow along the tangential direction of the droplet surface, while the growth direction of the dendrites in the hole is more random for the complex heat flow in the hole zone.

For hypoeutectic $Zr_{64}V_{36}$ alloy, the surface morphology of the sample with an undercooling of 253 K (**Figure 5a**) shows a hole-like shrinkage cavity. The metastable liquid state with undercoolings up to 87 and 253 K have been achieved for liquid refractory Nb—Si and Zr—V alloys in outer space. It provides the possibility for exploring the fancy solidification phenomena under microgravity conditions. We can find that the surface of $Zr_{64}V_{36}$ alloy (Figure 5b) is smoother compared to Figure 1b. Figure 5c shows the internal morphology of the shrinkage cavity, and the depth of the shrinkage cavity is estimated to be ≈0.5 mm, which is lower than half of the shrinkage cavity A in $Nb_{82.7}Si_{17.3}$ alloy. The overall morphology of the shrinkage cavity and some characteristic microstructures are marked by red boxes in Figure 5d. Figure 5e,f shows some significant dendrite microstructures on the inner wall: dendrites with no secondary arms and dendrites growing perpendicular to the surface. Compared with the surface morphology of eutectic $Nb_{82.7}Si_{17.3}$ alloy, there is only one shrinkage cavity on the surface of $Zr_{64}V_{36}$ alloy. The primary (Zr) phase of hypoeutectic $Zr_{64}V_{36}$ alloy shows an irregular strip organization without secondary dendrite arms (Figure 5d,e). It can be noted that the tip profile of (Zr) dendrite slightly deviates from the ideal paraboloid dendrite tip (Figure 5g, goodness-of-fit value is 0.957). Usually, the growth shape of solid solution phases from liquid is dendrite, resulting in a smooth interface at the tip of it, as shown in Figure 5f,h. The (Zr) phase exhibits a BCC crystal

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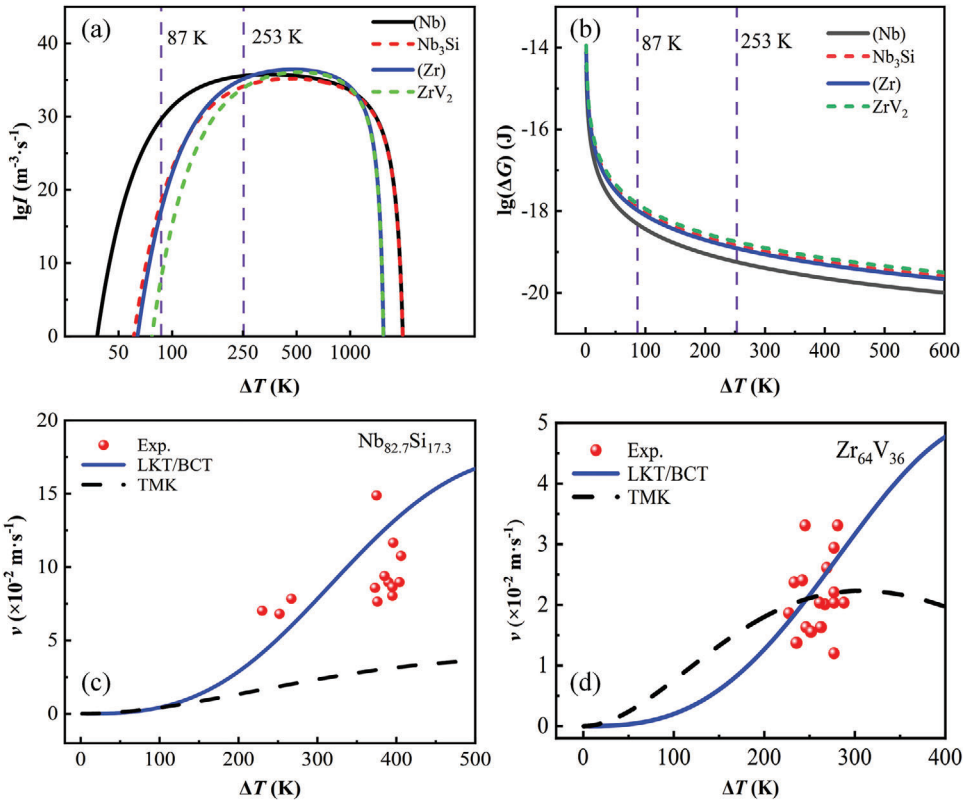

Figure 6. Nucleation rate and growth velocity of eutectic $Nb_{82.7}Si_{17.3}$ alloy and hypoeutectic $Zr_{64}V_{36}$ alloy. a,b) Heterogeneous nucleation rate and critical work. c,d) Crystal growth velocity.

structure upon solidification at a high temperature, whose preferential growth direction is [001] direction. When the growth velocity of the (110) plane is less than the (100) plane, the cross-section profile of the dendrite will be like the four-pointed star (Figure 5h). Figure 5i displays the space geometrical relationship of the (100) and (110) planes. The formation of a polyhedral structure at the dendrite tip of (Zr), as depicted in Figure 5f,g, can be attributed to the solid–solid phase transition from BCC to hexagonal close-packed crystal structures. Since both the (Zr) phase and (Nb) phase are BCC lattice structures at a high temperature, why does the morphology of primary (Zr) not display the typical dendrite with secondary arms? A reasonable explanation is that the eutectic growth inhibits the secondary arms growth of the primary phase (see the next Section).

2.3. Synergistic Effects of Crystal Nucleation and Growth with Shrinkage Cavity Formation

Due to the combination of microgravity and high undercooling, it is necessary to analyze the nucleation and dendritic growth dynamics. The wetting factor of $Nb_{82.7}Si_{17.3}$ alloy is chosen to 0.01,[20] and the same value is chosen for $Zr_{64}V_{36}$ alloy. Other required parameters can be found in the literature.[20,34–36] The nucleation rate and critical nucleation work of (Zr) and V_2Zr are close when the undercooling is 253 K, which implies that the coupled growth of primary dendrites and eutectics is more likely to occur within $Zr_{64}V_{36}$ alloy (**Figure 6**a,b). When the undercooling is not particularly small, the growth velocity under the microgravity of 10^{-5} g_0 is not much different from it under normal gravity.[37] As for $Nb_{82.7}Si_{17.3}$ alloy, the value calculated by the LKT/BCT model is used to estimate the growth velocity of the primary (Nb) phase, which is 3.2 mm s^{-1} when the undercooling is 87 K (Figure 6c). The temperature–time curve of hypoeutectic $Zr_{64}V_{36}$ alloy (Figure 5a) exhibits a single recalescence peak, indicating the simultaneous growth of primary dendrites and eutectics at this undercooling. The calculated result in Figure 6d confirms that their growth velocities are closely matched, supporting the aforementioned inference. Consequently, the primary and eutectic phases undergo coupled growth, resulting in a consistent growth direction (Figure 5d). Combining the experimental data with the calculated data, it is determined that the growth velocity of the primary (Zr) phase at 253 K undercooling is ≈21.7 mm s^{-1}.

Comparing eutectic $Nb_{82.7}Si_{17.3}$ alloy whose undercooling is 87 K with hypoeutectic $Zr_{64}V_{36}$ alloy whose undercooling is 253 K, the former has lower nucleation rate and larger critical nucleation work, which means fewer nucleations would appear at the same time. However, according to the experimental results aboard CSS, there were more nucleation sites in $Nb_{82.7}Si_{17.3}$ alloy with an undercooling of 87 K. The growth velocity can be used to explain this phenomenon. The freezing velocity of $Nb_{82.7}Si_{17.3}$ alloy is only one-seventh that of $Zr_{64}V_{36}$ alloy, which means the $Nb_{82.7}Si_{17.3}$ alloy droplet has 7 times the growth time, the difference in growth time is sufficient to compensate for the nucleation rate and nucleation work, which results in multiple nucleation sites on the surface of eutectic $Nb_{82.7}Si_{17.3}$ alloy. This indicates

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that if an alloy droplet has a high nucleation rate, a low critical nucleation work, and meanwhile slower growth velocity, it will have more groove-like shrinkage cavities on its surface after solidification. We conclusively deduce that microgravity solidification dynamics leads to obvious freezing shrinkages, and the crystal growth velocity and nucleation are the dominant factors. Therefore, what causes the difference in surface morphology between $Nb_{82.7}Si_{17.3}$ and $Zr_{64}V_{36}$ alloys?

Based on the previous results, nucleation should be regarded as the most important factor. As shown in the FEM simulation result (Figure 2f), in the solidification process, the flow velocity at the surface of the droplet is greater than that in the interior. Therefore, nucleation is likely to appear on the droplet surface. The primary phase is formed simultaneously or successively in different areas on the surface of the eutectic $Nb_{82.7}Si_{17.3}$ alloy droplet, and then the residual liquid solidifies radially outward with the nucleation sites as the centers. Since the remaining liquid is not enough to fill the whole region, shrinkage cavities are generated at the junction of these nucleation regions, where the surface effect plays a more pronounced role under microgravity. For the hypoeutectic $Zr_{64}V_{36}$ alloy, the primary phase and eutectic phase form simultaneously and their growth velocity is close at such an undercooling, which lead to the coupled growth. The outer layer dissipates heat faster, so its liquid–solid interface advances slightly faster, and a regular shrinkage cavity is formed on the opposite side of the nucleation site. In addition, after the beginning of solidification, with the release of the latent heat of solidification, the temperature at the front edge of the liquid–solid interface will rise rapidly, the magnitude of flow velocity can be inferred to the order of 10^{-3} m s^{-1} from the FEM simulation in Figure 2f. The asymmetry of dendrites in Figure 3 and Figure 4 should be attributed to the large flow velocity.[38] This indicates that outer space provides an ideal environment of dendritic growth as its nature, especially on a liquid surface, which is hard to achieve on the ground.

From the above analyses, the microgravity experiments suggest that novel microstructure patterns are revealed, such as dendrites growing along the tangential and normal directions to the droplet surface, the cavity forming at any location in the droplet surface, the coupled growth of primary and eutectic phases.

3. Conclusion

We reveal the freezing shrinkage dynamics and surface dendritic growth modes of floating refractory alloy droplets in outer space. The highest temperature up to 2265 and 1800 K together with 87 and 253 K undercoolings deviating from thermodynamically stable state were attained for $Nb_{82.7}Si_{17.3}$ and $Zr_{64}V_{36}$ refractory alloys, respectively. If two growing crystal grains originated from two independent nucleation sites meet together, then the solidification of residual liquids at the junctions usually engenders a groove-like cavity. Meanwhile, the hole-like cavity often appears at the intersection of growing grains from three or more nucleation sites. The calculation shows that the difference in microstructure patterns between the two alloys is mainly affected by nucleation rate and growth velocity. Furthermore, we have found that primary dendrites grow not only along the tangential direction but also along the normal direction to the droplet surface, which results from the effects of shrinkage dynamics and liquid surface flow. The results under the condition of microgravity contribute to a further understanding of the solidification behavior of undercooled liquid alloys under a weaker convection condition.

4. Experimental Section

Materials Preparations: The samples used in the containerless experiment were eutectic $Nb_{82.7}Si_{17.3}$ alloy, whose eutectic plateau temperature is 2189 K. The samples were made of pure Nb (99.95% purity) and pure Si (99.999% purity) by laser heating and melting under an Ar atmosphere, and the diameter of the spherical samples is ≈2.7 mm. The samples of hypoeutectic $Zr_{64}V_{36}$ alloy were prepared from pure Zr (99.995% purity) and pure V (99.95% purity) with the same preparation steps as that of eutectic $Nb_{82.7}Si_{17.3}$ alloy. According to the test, the mass loss of the eutectic $Nb_{82.7}Si_{17.3}$ alloy sample and hypoeutectic $Zr_{64}V_{36}$ alloy sample is less than 0.5% after being superheated for 100 K under the vacuum condition of 10^{-4} Pa for 10 min. Therefore, such mere mass loss and composition deviation could be ignored during the calculation.

ESL Experiment in Outer Space: The containerless material experiment facility located aboard CSS controls the samples by means of electrostatic levitation, and the suspended sample was heated by four heating laser beams located at the vertices of the regular tetrahedron. The four heating laser beams were switched on and off simultaneously and maintained the same power to reduce the influence of light pressure on the sample position. During the experiment, the pressure inside the experimental cavity was maintained at the order of 10^{-4} Pa. The eutectic $Nb_{82.7}Si_{17.3}$ alloy was kept at a superheated temperature of ≈100 K for tens of seconds, and then all the heating laser beams were turned off. After that, the sample was cooled by natural radiation. The first recalescence occurred with an undercooling of 90 K after a few seconds, then the temperature continued to drop to ≈2000 K for the second recalescence and remained at the eutectic plateau temperature for several seconds until it was completely solidified. For the hypoeutectic $Zr_{64}V_{36}$ alloy, its heating process was like that of the eutectic $Nb_{82.7}Si_{17.3}$ alloy. However, the hole-like shrinkage cavity appeared on the surface only when the undercooling was 253 K with only one recalescence, while there were no shrinkage cavities when the undercooling was small.

Microstructure Characterization: Scanning electron microscope micrographs of impact sites were acquired using a FEI Verios G4 field emission SEM at 15 kV operating voltage. CT scan photographs of spherical samples with a diameter of ≈2.7 mm were obtained by a Zeiss Xradia 620 Versa 3D X-ray microscope at 120 kV operating voltage.

Temperature and Flow Field Simulation: The finite element method was used for numerical simulation. It was not necessary to consider gravity when conducting containerless experiments on the space station. The N–S equation and the continuity equation in this study can be expressed as:[39]

$$\nabla \cdot \boldsymbol{u} = 0 \tag{1}$$

$$-\nabla \boldsymbol{p} + \nabla \cdot \left(\mu \left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T\right)\right) = 0 \tag{2}$$

where $\boldsymbol{u}$ is the fluid velocity, $\boldsymbol{p}$ is the fluid pressure, and μ is the fluid viscosity. Fluid heat transfer is described by the heat transfer equation and Fourier's law of heat transfer:[40,41]

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \boldsymbol{u} \cdot \nabla T + \nabla \cdot \boldsymbol{q} = Q + Q_p + Q_{vd} \tag{3}$$

$$\boldsymbol{q} = -\kappa \nabla T \tag{4}$$

where κ is the thermal conductivity of the liquid, $\boldsymbol{q}$ is the heat flux and C_p is the specific heat at constant pressure. Q is the external heat source, Q_p is the pressure work and Q_{vd} is the viscous heat dissipation. In this study, both Q_p and Q_{vd} can be ignored. Under microgravity conditions, the Marangoni effect at the interface should be considered for fluid flow:

$$\sigma_t = \left|\frac{\partial \sigma}{\partial T}\right| \vec{\nabla} T \cdot \boldsymbol{t} \tag{5}$$

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where σ_t is the surface tension, $\vec{\nabla}T$ is the temperature gradient along the surface, and $\boldsymbol{t}$ is the tangential unit vector which is parallel to the surface temperature gradient. For incompressible fluid, the velocity change caused by the Marangoni effect at the interface is as follows:

$$\left[\mu\left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T\right)\right] \cdot \boldsymbol{n} = \sigma_t \tag{6}$$

According to the actual situation of the experiment aboard CSS, a 3D sphere model with a diameter of 2.7 mm was established, and the inward heat flux on the surface was added at the four vertices of the regular tetrahedron to simulate the actual heating laser. The heating time and the maximum heating temperature were set according to the experimental process. All calculations were completed using COMSOL Multiphysics software.

Phase Transition Simulation: The apparent heat capacity method was used to simulate the phase transition process. The heat capacity is redefined with the addition of latent heat:[42,43]

$$C_p = \frac{1}{\rho}\left(\theta_1\rho_1 C_{p,1} + \theta_2\rho_2 C_{p,2}\right) + L\frac{\partial \alpha_m}{\partial T} \tag{7}$$

$$\rho = \theta_1\rho_1 + \theta_2\rho_2 \tag{8}$$

$$\alpha_m = \frac{1}{2}\frac{\theta_2\rho_2 - \theta_1\rho_1}{\theta_1\rho_1 + \theta_2\rho_2} \tag{9}$$

ρ_1 and ρ_2 are the densities, here phase 1 is chosen as liquid and phase 2 as solid, C_p is the specific heat, L is the latent heat, θ is the phase field order parameter and it follows the relationship: $\theta_1 + \theta_2 = 1$. The Heaviside function is chosen to describe the change of θ when the temperature changes, a common expression is as follows:

$$H(x) = \theta_1 = \frac{1}{2}\left[1 + \frac{a(T - T_m)}{(1 + a^2(T - T_m)^2)^{\frac{1}{2}}}\right] \tag{10}$$

where a is the coefficient decided by the temperature transition interval δT. An additional source item F is added in the left of Equation (2) to dampen the velocity at the solid–liquid interface so that it becomes the solidified phase after the transition:[42,44]

$$\boldsymbol{F} = \frac{(1 - f_L)^2}{f_L^{\,3} + \varepsilon} A\boldsymbol{u} \tag{11}$$

f_L is the real-time volume fraction of the liquid phase, A represents an arbitrary large constant and ϵ represents a small constant which would produce a proper damping. A 2D circle model with a diameter of 2.7 mm was established to show the phase transition process.

Crystal Growth Kinetics Calculation: The critical work ΔG of the heterogeneous nucleation and the nucleation rate I describe the ease of nucleation from liquid.[20,45] The growth velocities of the primary phase and the eutectic phase were numerically calculated by LKT/BCT dendrite growth model[46–49] and TMK eutectic growth model,[50,51] respectively.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

dendritic growth, microgravity, phase transition, refractory alloys, undercooling

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